Funktionen der Treiber-DLL “MAIv2DLL.dll”

About

This Helpfile describes the abstract programming interface implemented in the File “*MAIv2DLL.dll*” for measurement hardware from the company Goldammer GmbH, Wolfsburg, Germany.

This DLL supports all measurement hardware from the series USB Basic, USB MC4 and PCI MC4. A complete overview over all available devices an addons can be found on the website Internetseite http://www.goldammer.de/.

This document was automatically generated from the DLL’s sourcecode at version ?.?.?



Funktionen der Treiber-DLL “MAIv2DLL.dll”



device pool functions.

Functions that apply to all devices are:

GetFirstDevice()

GetNumberOfDevices()

GetDeviceHandles()

GetDeviceHandlesMultichoiceETH()

GetDeviceHandlesMultichoicePCI()

GetDeviceHandlesMultichoiceUSB()

GetDeviceHandlesMultichoiceUSBBasic()

GetFirstDeviceByType()

OpenDevice()

CloseDevice()

IsDeviceOpen()

GetErrorDescription()

SetGPCIFile()

GetGPCIFilePath()

SetMAIDevicelistChangedHandler()

ReCreateDeviceListOmittingDeviceTypes()

ShowMessageBoxOnDeviceCreationError()

GetNumberOfUninitializedDevices()

GetDeviceCreationErrorMessage()

DeviceInfoGet()

DeviceInfoFree()

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CloseDevice

```
long CloseDevice( uint nInstance );
```

obsolete function.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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DeviceInfoFree

```
long DeviceInfoFree( structmaidevinfo* info );
```

Return handles of all available devices

info	array pointer to
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long DeviceInfoGet( enumeDeviceClass type,
structmaidevinfo** info );
```

type	…
info	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long GetDeviceCreationErrorMessage( uint
nNumber, constchar[] AOText, uint
nTextBufferLength );
```

nNumber	…
AOText	…
nTextBufferLength	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetDeviceHandles



```
long GetDeviceHandles( uint[] Handles, uint*
iHandleCount );
```

Return handles of all available devices

Handles	array pointer to store all the device handles
iHandleCount	maximum count of handles to copy
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetDeviceHandlesMultichoiceETH



```
long GetDeviceHandlesMultichoiceETH( uint[] Handles, uint* iHandleCount );
```

Return handles of all available multichoice eth devices

Handles	array pointer to store all the device handles
iHandleCount	maximum count of handles to copy
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetDeviceHandlesMultichoicePCI

```
long GetDeviceHandlesMultichoicePCI( uint[] Handles, uint* iHandleCount );
```

Return handles of all available multichoice pci devices

Handles	array pointer to store all the device handles
iHandleCount	maximum count of handles to copy
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetDeviceHandlesMultichoiceUSB



```
long GetDeviceHandlesMultichoiceUSB( uint[] Handles, uint* iHandleCount );
```

Return handles of all available multichoice usb devices

Handles	array pointer to store all the device handles
iHandleCount	maximum count of handles to copy
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetDeviceHandlesMultichoiceUSBBasic

```
long GetDeviceHandlesMultichoiceUSBBasic( uint[] Handles, uint* iHandleCount );
```

Return handles of all available multichoice usb basic devices

Handles	array pointer to store all the device handles
iHandleCount	maximum count of handles to copy
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long GetErrorDescription( uint nInstance, long
ErrorNumber, char[] ErrorDescription );
```

returns a desctiption text for an error number.

nInstance	index of the device.
ErrorNumber	error code.
ErrorDescription	[out] desctiption text.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long GetFirstDevice( );
```

returns the index of the first device.

return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetFirstDeviceByType

```
long GetFirstDeviceByType( uint DeviceType );
```

returns the index of the first device of a certain type.

DeviceType	device type to find.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetGPCIFilePath



```
const char * GetGPCIFilePath( );
```

return value resulting string.

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GetNumberOfDevices



```
long GetNumberOfDevices( );
```

return value If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetNumberOfUninitializedDevices



```
long GetNumberOfUninitializedDevices( );
```

return value If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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IsDeviceOpen

```
long IsDeviceOpen( uint nInstance );
```

obsolete function.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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OpenDevice

```
long OpenDevice( uint nInstance );
```

obsolete function.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long ReCreateDeviceListOmittingDeviceTypes( bool
AICreatePCI, bool AICreateUSBMC4, bool
CreateUSBBasic );
```

AICreatePCI	…
AICreateUSBMC4	…
CreateUSBBasic	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long SetGPCIFile( constchar[] AIFilePath, uint
nLength );
```

sets the Filename (or path) of the GPCI file used for initializing PCI cards.

AIFilePath	Filename (or path) of the GPCI file.
nLength	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long SetMAIDevicelistChangedHandler(
MAI_DEVICELISTCHANGED_CALLBACK
AIMAIDevicelistChangedFunction );
```

AIMAIDevicelistChangedFunction …

return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long ShowMessageBoxOnDeviceCreationError( bool
AIShowMsgBox );
```

AIShowMsgBox …

return value If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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Funktionen der Treiber-DLL “MAIv2DLL.dll”



device functions.

for controlling the devices non channel dependent features these functions are available:

ClearAllChannelLists()

ResetDevice()

LoadFirmware()

StartMeasure()

StopMeasure()

SetStreamingMode()

SetDataBufferModes()

SetRunningMode()

SetRunningModeEx()

EnableSoftwareClock()

ConfigCard()

SetGlobalTriggerValues()

ConfigMeasure()

IdentifierDevice()

IsUSBHighSpeed()

RemoteDeviceConnect()

RemoteDeviceDisconnect()

RemoteDevicesSearch()

RemoteDevicesSearchAndConnectAll()

RemoteDevicesGetIPFromSearchResult()

RemoteDeviceGetIPFromPoolHandle()

GetCardName()

GetCardProducer()

GetCardSeller()

GetManufacture_Name()

GetSerialNumber()

GetCustomer_Name()

GetCustomer_Number()

GetCustomer_DateOfPurchase()

GetADChannels()

GetADChannelsNumberOfOversamplingModes()

GetADChannelsNumberOfGainModes()

GetADSampleWidth()

GetADSampleRate()
GetDAChannels()
GetDASampleWidth()
GetDASampleRate()
GetTTLChannels()
GetTTLSampleWidth()
GetTTLSampleRate()
GetCTChannels()
GetCTSampleWidth()
GetCTSampleRate()
GetPWMChannels()
GetPWMSampleWidth()
GetPWMSampleRate()
GetSoftwareExtension()
GetHardwareExtension()
GetDevicePropertyStringA()
GetMeasurementProcessorVersion()
GetControllerFirmwareVersion()
GetSystemDriverVersion()
PrivateFlashDataRead()
PrivateFlashDataWrite()
PrivateFlashDataClear()
SetLEDs()
GetDLLVersion()
GetLIBVersion()
GetCardIDBySerial()
GetDeviceInfo()
GetNumberOfBurstsOnADController()
GetNumberOfBurstsOnCTController()
AlteraRegisterRead()
AlteraRegisterWrite()

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```
long AlteraRegisterRead( uint nCardID, int
nAdress, uint* nRegisterValue );
```

nCardID	…
nAdress	…
nRegisterValue	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AlteraRegisterWrite( uint nCardID, int
nAdress, uint nRegisterValue );
```

nCardID	…
nAdress	…
nRegisterValue	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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ClearAllChannelLists

```
long ClearAllChannelLists( uint nInstance );
```

clears all measurement channel lists.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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ConfigCard

```
long ConfigCard( uint nInstance, uint
TriggerExtern, uint ExternEdge, uint
DifferenzMode, uint p4 );
```

configures the external triggering setup.

nInstance	index of the device.
TriggerExtern	external trigger mode.
ExternEdge	external trigger edge.
DifferenzMode	toggles differential measurement mode.
p4	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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ConfigMeasure



```
long ConfigMeasure( uint nInstance );
```

configures the hardware for a measurement.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long EnableSoftwareClock( uint nInstance, uint
AIFetchADDataInUserMode,
uintAISoftwareADBufferSizeInBursts =1 );
```

toggles the software AD mode.

nInstance	index of the device.
AIFetchADDataInUserMode	0 = Software AD mode is OFF.
=1	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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Funktionen der Treiber-DLL “MAIv2DLL.dll”

GetADChannels



```
long GetADChannels( uint nInstance );
```

returns the nuber of AD Channels.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long GetADChannelsNumberOfGainModes( uint
nInstance );
```

nInstance	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long GetADChannelsNumberOfOversamplingModes(
uint nInstance );
```

nInstance	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetADSampleRate

```
long GetADSampleRate( uint nInstance );
```

returns the maximum sample rate of the AD Channels.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetADSampleWidth

```
long GetADSampleWidth( uint nInstance );
```

returns the sample width of the AD Channels.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetCardIDBySerial



```
long GetCardIDBySerial( char[] Serial, uint* nCardID );
```

gets the ID of a Card.

Serial	[in] serialnumber of the device.
nCardID	[out] index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetCardName

```
const char * GetCardName( uint nInstance );
```

returns the name of this device.

nInstance	index of the device.
return value	resulting string.

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GetCardProducer

```
const char * GetCardProducer( uint nInstance );
```

returns the name of the card producer.

nInstance	index of the device.
return value	resulting string.

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GetCardSeller

```
const char * GetCardSeller( uint nInstance );
```

returns the name of the card vendor.

nInstance	index of the device.
return value	resulting string.

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```
long GetControllerFirmwareVersion( uint
nInstance );
```

returns version nuber of the firmware.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetCTChannels

```
long GetCTChannels( uint nInstance );
```

returns the nuber of CT Channels.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetCTSampleRate

```
long GetCTSampleRate( uint nInstance );
```

returns the maximum sample rate of the CT Channels.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long GetCTSampleWidth( uint nInstance );
```

returns the sample width of the CT Channels.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetCustomer_DateOfPurchase

```
const char * GetCustomer_DateOfPurchase( uint
nInstance );
```

returns the date of purchase.

nInstance	index of the device.
return value	resulting string.

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GetCustomer_Name

```
const char * GetCustomer_Name( uint nInstance );
```

returns the name of the customer.

nInstance	index of the device.
return value	resulting string.

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GetCustomer_Number

```
const char * GetCustomer_Number( uint nInstance
);
```

returns the unique number of the customer.

nInstance	index of the device.
return value	resulting string.

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GetDAChannels

```
long GetDAChannels( uint nInstance );
```

returns the nuber of DA Channels.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetDASampleRate

```
long GetDASampleRate( uint nInstance );
```

returns the maximum sample rate of the DA Channels.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetDASampleWidth

```
long GetDASampleWidth( uint nInstance );
```

returns the sample width of the DA Channels.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetDeviceInfo

```
long GetDeviceInfo( uint nCardID, int*
nDeviceType, char[] Serial );
```

returns serial number and devicetype.

nCardID	…
nDeviceType	[out] type of the device.
Serial	[out] serialnumber of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetDevicePropertyStringA

```
long GetDevicePropertyStringA( uint nInstance,
constchar* pszPropertyName, char* pData, uint*
pcbData );
```

Retreive a property value stored by the mai measurement device.

nInstance	[in] index of the device.
pszPropertyName	[in] Pointer to a null-terminated string that contains the name of the property to request.
pData	[out] A caller-allocated buffer to hold the retrieved data.
pcbData	[in,out] The size of the buffer allocated or returned by pData, in bytes.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetDLLVersion

```
const char * GetDLLVersion( );
```

returns version number of the MaiV2DLL.

return value resulting string.

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```
long GetHardwareExtension( uint nInstance, uint*
pHWExtension );
```

returns the installed hardware extensions (ORed in one DWORD).

nInstance	index of the device.
pHWExtension	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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Funktionen der Treiber-DLL “MAIv2DLL.dll”



GetLIBVersion

```
const char * GetLIBVersion( );
```

returns version number of the internal MAIv2LIB.

return value resulting string.

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GetManufacture_Name

```
const char * GetManufacture_Name( uint nInstance
);
```

returns the name of the card’s board.

nInstance	index of the device.
return value	resulting string.

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```
long GetMeasurementProcessorVersion( uint
nInstance );
```

returns version nuber of the internal measurement processor.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long GetNumberOfBurstsOnADController( uint
nCardID, uint* nNumberOfBursts );
```

nCardID	…
nNumberOfBursts	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long GetNumberOfBurstsOnCTController( uint
nCardID, uint* nNumberOfBursts );
```

nCardID	…
nNumberOfBursts	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetPWMChannels

```
long GetPWMChannels( uint nInstance );
```

returns the nuber of PWM Channels.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetPWMSampleRate

```
long GetPWMSampleRate( uint nInstance );
```

returns the maximum sample rate of the PWM Channels.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetPWMSampleWidth

```
long GetPWMSampleWidth( uint nInstance );
```

returns the sample width of the PWM Channels.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetSerialNumber

```
const char * GetSerialNumber( uint nInstance );
```

returns the serial number of the card’s board.

nInstance	index of the device.
return value	resulting string.

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```
long GetSoftwareExtension( uint nInstance, uint*
pSWExtension );
```

returns the installed software extensions (ORed in one DWORD).

nInstance	index of the device.
pSWExtension	[out] installed software extensions.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long GetSystemDriverVersion( uint nInstance,
char[] DLLVersion );
```

returns the version of the WDM system driver.

nInstance	index of the device.
DLLVersion	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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GetTTLChannels

```
long GetTTLChannels( uint nInstance );
```

returns the nuber of TTL Channels.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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Funktionen der Treiber-DLL “MAIv2DLL.dll”

GetTTLSampleRate



```
long GetTTLSampleRate( uint nInstance );
```

returns the maximum sample rate of the TTL Channels.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long GetTTLSampleWidth( uint nInstance );
```

returns the sample width of the TTL Channels.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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IdentifierDevice

```
long IdentifierDevice( uint nInstance );
```

nInstance	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long IsUSBHighSpeed( uint nInstance, uint*
IsUSBHighSpeed );
```

nInstance	…
IsUSBHighSpeed	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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LoadFirmware

```
long LoadFirmware( uint nInstance );
```

resets the device and reloads the firmware if necessary.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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PrivateFlashDataClear

```
long PrivateFlashDataClear( uint nInstance );
```

clears the private flash memory.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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PrivateFlashDataRead

```
long PrivateFlashDataRead( uint nInstance,
uchar[] PrivateData, uint nLength );
```

reads the private flash memory.

nInstance	index of the device.
PrivateData	[out] private flash memory data
nLength	number of bytes
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long PrivateFlashDataWrite( uint nInstance,
uchar[] PrivateData, uint nLength );
```

writes into the private flash memory.

nInstance	index of the device.
PrivateData	private flash memory data
nLength	number of bytes
return value	If no error occurs the value "*0*" is returned, otherwise a sub-zero error code.

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RemoteDeviceConnect

```
long RemoteDeviceConnect( uint* nInstance, char*
sIPAdress, int nUDPINPort, bool
bShowErrorMessage, bool* bIsNewInDevicelist );
```

nInstance	…
sIPAdress	…
nUDPINPort	…
bShowErrorMessage	…
bIsNewInDevicelist	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long RemoteDeviceDisconnect( uint nInstance );
```

nInstance	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long RemoteDeviceGetIPFromPoolHandle( uint
nInstance, char[] sIPv4Address, int nBuffersize,
uint* nTCPPort, int* nUDPPort );
```

nInstance	…
sIPv4Address	…
nBuffersize	…
nTCPPort	…
nUDPPort	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long RemoteDevicesGetIPFromSearchResult( int
nIndex, char[] sIPv4Address, int nBuffersize );
```

nIndex	…
sIPv4Address	…
nBuffersize	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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RemoteDevicesSearch

```
long RemoteDevicesSearch( );
```

return value If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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RemoteDevicesSearchAndConnectAll

```
long RemoteDevicesSearchAndConnectAll( );
```

return value If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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ResetDevice

```
long ResetDevice( uint nInstance );
```

resets the device .

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long SetDataBufferModes( uint nInstance, uint
bGetADDataInStreamBuffer, uint
bGetCTDataInStreamBuffer, uint
bGetTTLDataInStreamBuffer );
```

nInstance	…
bGetADDataInStreamBuffer	…
bGetCTDataInStreamBuffer	…
bGetTTLDataInStreamBuffer	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long SetGlobalTriggerValues( uint nInstance,
uint nPreTrigger, uint nPostTrigger, uint
nWaitForStartTrigger );
```

Sets the pre and post trigger values.

nInstance	index of the device.
nPreTrigger	number of values before trigger events to keep (must be smaller than the size of the cards channelbuffer).
nPostTrigger	number of values after trigger events to keep. (must be smaller than the size of the cards channelbuffer).
nWaitForStartTrigger	determines, if the recording of values starts immediately when StartMeasure is called or if a start trigger is used.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long SetLEDs( uint nInstance, bool
AIRedOrFirstLEDOn, bool AIGreenOrSecondLEDOn,
bool AIThirdLEDOn );
```

nInstance	…
AIRedOrFirstLEDOn	…
AIGreenOrSecondLEDOn	…
AIThirdLEDOn	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long SetRunningMode( uint nInstance, uint
MasterSlave );
```

sets the running mode (master/slave/standalone).
ATTENTION. If AIAllowMultipleMasters is set to [true], the connetor on ALL devices is set to HIGH on measurement start. This can damage/destroy the hardware if these pins of to master cards are connected !!!

nInstance	index of the device.
MasterSlave	running mode (master=1/slave=2/standalone=0).
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long SetRunningModeEx( uint nInstance, uint
MasterSlave, bool AIMultipleMastersAllowed );
```

sets the running mode (master/slave/standalone).
ATTENTION. If AIAllowMultipleMasters is set to [true], the connetor on ALL devices is set to HIGH on measurement start. This can damage/destroy the hardware if these pins of to master cards are connected !!!

nInstance	index of the device.
MasterSlave	running mode (master=1/slave=2/standalone=0).
AIMultipleMastersAllowed	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long SetStreamingMode( uint nInstance, uint
nStreamingMode );
```

toggles the streaming mode.

nInstance	index of the device.
nStreamingMode	0 = streaming mode is OFF.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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StartMeasure

```
long StartMeasure( uint nInstance );
```

starts all measurements that are configured.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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StopMeasure

```
long StopMeasure( uint nInstance );
```

stops all measurements that are running.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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Funktionen der Treiber-DLL “MAIv2DLL.dll”



AD functions.

for controlling and measuring AD Channels these functions are available:

AD_ClearChannelList()

AD_AddChannelToList()

AD_GetChannelListSize()

AD_ReadSingleValue()

AD_ReadSingleValueExt()

AD_ReadSingleVoltage()

AD_SetFrequency()

AD_GetFrequency()

AD_ReadLastBurst()

AD_ReadData()

AD_ReadDataExt()

AD_ReadBinaryData()

AD_GetBinaryWordsizePerChannel()

AD_ConvertBinaryBurstToDoublePerChannel()

AD_GetStreamPipeLevel()

AD_GetNumberOfValues()

AD_StartMeasure()

AD_StopMeasure()

AD_SetIEPE()

AD_SetIEPEOnAudioModul()

AD_SetEventToBeRaisedWhenNumberOfBurstsAvailable()

AD_Audio_SetHPFilter()

AD_Audio_SetHPFilterResponse()

AD_ResetIEPE()

AD_GetIEPEType()

AD_GetCorrectionValue()

AD_SetCorrectionValue()

AD_WriteCorrectionValuesToDevice()

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AD_AddChannelToList

```
long AD_AddChannelToList( uint nInstance, uint
nChannel, uint nOversampling, uint nGain, uint
nBipolar );
```

adds an AD measurement channel.

nInstance	index of the device.
nChannel	index of the channel.
nOversampling	oversampling mode.
nGain	gain.
nBipolar	toggles polarity .
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_Audio_SetHPFilter( uint nInstance, uint
nChannel, uint bFilterOn );
```

toggles the high pass filter on a G0V card.

nInstance	index of the device.
nChannel	index of the channel.
bFilterOn	0 = filter is off, 1 = filter is on.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_Audio_SetHPFilterResponse( uint
nInstance, uint nADCNr, uint
bLowGroupDelayResponseOn );
```

toggles the group delay for the high pass filter on a G0V card.

nInstance	index of the device.
nADCNr	index of the channel.
bLowGroupDelayResponseOn	0 = classic group delay, 1 = low group delay.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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AD_ClearChannelList

```
long AD_ClearChannelList( uint nInstance );
```

clears the AD measurement channel list.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_ConvertBinaryBurstToDoublePerChannel(
uint nInstance, uchar[] BinaryData, double*
nConvertedBurst );
```

nInstance	…
BinaryData	…
nConvertedBurst	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_GetBinaryWordsizePerChannel( uint
nInstance, uint* nSingleADWordSize );
```

nInstance	…
nSingleADWordSize	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_GetChannelListSize( uint nInstance,
uint* nCLSize );
```

returns the length of the AD measurement channel list.

nInstance	index of the device.
nCLSize	[out] AD measuement channel count.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_GetCorrectionValue( uint nInstance, uint
nChannel, float* fValue );
```

reads the [out] correction value for one analog channel.

nInstance	index of the device.
nChannel	index of the channel.
fValue	[out] correction value.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_GetFrequency( uint nInstance, double* fFrequency );
```

returns the AD samplerate.

nInstance	index of the device.
fFrequency	[out] AD samplerate in Hz.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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AD_GetIEPEType

```
long AD_GetIEPEType( uint nInstance );
```

nInstance	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_GetNumberOfValues( uint nInstance, uint*
nNumberOfValues );
```

returns number of measured bursts.

nInstance	index of the device.
nNumberOfValues	[out] number of bursts.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_GetStreamPipeLevel( uint nInstance,
double* fStreamPipeLevel );
```

returns fifo fill level.

nInstance	index of the device.
fStreamPipeLevel	[out] fifo fill level in percent.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_ReadBinaryData( uint nInstance, uchar[] BinaryData, uint* nLength );
```

returns measure data in binary format.

nInstance	index of the device.
BinaryData	[out] measuredata in binary format.
nLength	[in,out] number of values (to) read.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_ReadData( uint nInstance, uint
nChannelListEntry, int[] RawData, double[]
Value, uint* nValuesPerChannel, uint bPartBlock,
uint bSkipData );
```

returns measure data in binary format.

nInstance	index of the device.
nChannelListEntry	index of the channel in the measurement channel list.
RawData	[out] raw data.
Value	[out] scaled data.
nValuesPerChannel	[in,out] number of values (to) read.
bPartBlock	toggles, if blocks smaller than nLength will be read.
bSkipData	obsolete.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_ReadDataExt( uint nInstance, uint
nChannelListEntry, int[] pRawData, double[]
pDoubleValues, float[] pFloatValues, uint*
nValuesPerChannel, uint bPartBlock, __int64*
AODateTimeT0, double* AODateTimeDeltaT );
```

returns measure data in binary format.

nInstance	index of the device.
nChannelListEntry	index of the channel in the measurement channel list.
pRawData	[out] raw data.
pDoubleValues	[out] scaled data as double array.
pFloatValues	[out] scaled data as float array.
nValuesPerChannel	[in,out] number of values (to) read.
bPartBlock	…
AODateTimeT0	T0 des ersten gelesenen Bursts.
AODateTimeDeltaT	delta t between the bursts.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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AD_ReadLastBurst

```
long AD_ReadLastBurst( uint nInstance, int[] RawData, double[] Value, uint* nNumberOfValues );
```

returns the last measured burst.

nInstance	index of the device.
RawData	[out] raw data.
Value	[out] scaled vlaues.
nNumberOfValues	[in,out] number of values (to) read.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_ReadSingleValue( uint nInstance, uint
nChannel, uint nOversampling, uint nGain, uint
nBipolar, int* nRawValue );
```

read single raw AD value.

nInstance	index of the device.
nChannel	index of the channel.
nOversampling	oversampling mode.
nGain	gain.
nBipolar	toggles polarity .
nRawValue	[out] raw value.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_ReadSingleValueExt( uint nInstance, uint
nChannel, uint nOversampling, uint nGain, uint
nBipolar, int* nRawValue, double* dVoltage,
float* fVoltage );
```

read single raw AD value.

nInstance	index of the device.
nChannel	index of the channel.
nOversampling	oversampling mode.
nGain	gain.
nBipolar	toggles polarity .
nRawValue	[out] raw value.
dVoltage	[out] voltage value as double.
fVoltage	[out] voltage value as float.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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AD_ReadSingleVoltage

```
long AD_ReadSingleVoltage( uint nInstance, uint
nChannel, uint nOversampling, uint nGain, uint
nBipolar, double* bVoltage );
```

read single AD Voltage.

nInstance	index of the device.
nChannel	index of the channel.
nOversampling	oversampling mode.
nGain	gain.
nBipolar	toggles polarity .
bVoltage	[out] Voltage.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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Funktionen der Treiber-DLL “MAIv2DLL.dll”

AD_ResetIEPE



```
long AD_ResetIEPE( uint nInstance );
```

restest all channel’s IEPE outout to the initial state..

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_SetCorrectionValue( uint nInstance, uint
nChannel, float fValue );
```

reads the [out] correction value for one analog channel (does NOT change the measurement values).

nInstance	index of the device.
nChannel	index of the channel.
fValue	correction value.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long
AD_SetEventToBeRaisedWhenNumberOfBurstsAvailable(
uint nInstance, void* hEventHandle, uint
nNumberOfBursts );
```

nInstance	…
hEventHandle	…
nNumberOfBursts	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_SetFrequency( uint nInstance, double
fFrequency );
```

sets the AD samplerate.

nInstance	index of the device.
fFrequency	AD samplerate in Hz.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_SetIEPE( uint nInstance, uint nChannel,
uint nIEPEMode, uint bGain );
```

sets the IEPE mode.

nInstance	index of the device.
nChannel	index of the channel.
nIEPEMode	IEPE mode.
bGain	if !=0, a 20dB Gain is applied.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_SetIEPEOnAudioModul( uint nInstance,
uint nChannel, uint bIEPEOn, uint bDCOn, uint
bCurrentSourceOn, uint bGainOn );
```

sets the IEPE mode on USB Basic audio moudules.

nInstance	index of the device.
nChannel	index of the channel.
bIEPEOn	IEPE mode on if != 0.
bDCOn	DC measurement, if !=0, otherwise AC.
bCurrentSourceOn	single ended measurement, if !=0, otherwise differential .
bGainOn	if !=0, a 20dB Gain is applied.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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AD_StartMeasure

```
long AD_StartMeasure( uint nInstance );
```

starts the AD measurement.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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AD_StopMeasure

```
long AD_StopMeasure( uint nInstance );
```

stops the AD measurement.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long AD_WriteCorrectionValuesToDevice( uint
nInstance );
```

writes the correction vlaues to the card’s flash memory.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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Funktionen der Treiber-DLL “MAIv2DLL.dll”



DA functions.

for controlling and measuring DA Channels these functions are available:

DA_SetRange()

DA_ClearChannelList()

DA_AddChannelToList()

DA_GetChannelListSize()

DA_SetFrequency()

DA_StartOutput()

DA_StopOutput()

DA_WriteChannelRaw()

DA_WriteChannelVoltage()

DA_WriteRawData()

DA_WriteScaledData()

DA_GetFifoState()

DA_Audio_SetMonauralMode()

DA_Audio_SetADLoopbackMode()

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```
long DA_AddChannelToList( uint nInstance, uint
nChannel );
```

adds an DA output channel.

nInstance	index of the device.
nChannel	index of the channel.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long DA_Audio_SetADLoopbackMode( uint nInstance,
uint bLoopbackModeOn );
```

nInstance	…
bLoopbackModeOn	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long DA_Audio_SetMonauralMode( uint nInstance,
uint bLoopbackModeOn );
```

nInstance	…
bLoopbackModeOn	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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DA_ClearChannelList

```
long DA_ClearChannelList( uint nInstance );
```

clears the DA measurement channel list.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long DA_GetChannelListSize( uint nInstance,
uint* nCLSize );
```

returns the length of the DA output channel list.

nInstance	index of the device.
nCLSize	[out] DA measuement channel count.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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DA_GetFifoState

```
long DA_GetFifoState( uint nInstance, uint
nChannel, uint* nNumberOfValues );
```

returns number of free space in bursts inside the output fifo.

nInstance	index of the device.
nChannel	…
nNumberOfValues	[out] free space in bursts.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long DA_SetFrequency( uint nInstance, double
fFrequency );
```

sets the DA output rate.

nInstance	index of the device.
fFrequency	DA output rate in Hz.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long DA_SetRange( uint nInstance, uint nRange );
```

sets the DA output range.

nInstance	index of the device.
nRange	0=Unipolar(0..10),1=Bipolar(-10..10).
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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DA_StartOutput

```
long DA_StartOutput( uint nInstance );
```

starts the DA output.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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DA_StopOutput

```
long DA_StopOutput( uint nInstance );
```

stops the DA output.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long DA_WriteChannelRaw( uint nInstance, uint
nChannel, int RawValue );
```

outputs a DA raw value.

nInstance	index of the device.
nChannel	index of the channel.
RawValue	raw value.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long DA_WriteChannelVoltage( uint nInstance,
uint nChannel, double VoltageValue );
```

outputs a DA voltage.

nInstance	index of the device.
nChannel	index of the channel.
VoltageValue	new voltage.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long DA_WriteRawData( uint nInstance, uint
nChannelListEntry, int[] pBuffer, uint* nLength,
uint bPartBlock );
```

returns measure data in binary format.

nInstance	index of the device.
nChannelListEntry	index of the channel in the measurement channel list.
pBuffer	…
nLength	[in,out] number of values to write/written.
bPartBlock	toggles, if blocks smaller than nLength will be written.
return value	If no error occurs the value "*0*" is returned, otherwise a sub-zero error code.

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```
long DA_WriteScaledData( uint nInstance, uint
nChannelListEntry, double[] fScaledValues, uint*
nLength, uint bPartBlock );
```

returns measure data in binary format.

nInstance	index of the device.
nChannelListEntry	index of the channel in the measurement channl list.
fScaledValues	Array of scaled values..
nLength	[in,out] number of values to write/written.
bPartBlock	toggles, if blocks smaller than nLength will be written.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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Funktionen der Treiber-DLL “MAIv2DLL.dll”



CT functions.

for controlling and measuring CT Channels these functions are available:

CT_ClearChannelList()

CT_AddChannelToList()

CT_GetChannelListSize()

CT_SetFrequency()

CT_GetFrequency()

CT_ScaleData()

CT_ReadData()

CT_ReadDataExt()

CT_ReadSingleValue()

CT_GetStreamPipeLevel()

CT_GetNumberOfValues()

CT_StartMeasure()

CT_StopMeasure()

CT_GetIncrementalCounterDirections()

CT_SetCounterParameter()

CT_GetCounterParameter()

CT_GetCounterParametersForMode()

CT_GetCounterParameterBoundaries()

CT_GetAvailableCounterModes()

CT_SetCounterMode()

CT_GetCounterMode()

CT_ImpulseCounter()

CT_FrequencyCounter()

CT_PeriodCounter()

CT_PulswidthCounter()

CT_IncrementalCounter()

CT_IncrementalCounterExtensionTime()

CT_IncrementalCounterExtensionFlow()

CT_UpDownCounter()

CT_SetCounterParameter_SingleValueMode()

CT_GetCounterParameter_SingleValueMode()

CT_SetCounterMode_SingleValueMode()

CT_GetCounterMode_SingleValueMode()

CT_Reset()

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```
long CT_AddChannelToList( uint nInstance, uint
nChannel, bool bToADChannellist );
```

adds an CT measurement channel.

nInstance	index of the device.
nChannel	index of the channel.
bToADChannellist	if [true] the channel will be added to the AD channellist (uses AD samplerate and measurement control functions).
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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CT_ClearChannelList

```
long CT_ClearChannelList( uint nInstance );
```

clears the CT measurement channel list.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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CT_FrequencyCounter

```
long CT_FrequencyCounter( uint nInstance, uint
nChannel, uint nResolution, uint nEdge, uint
nEachPeriod );
```

configures this channel as an frequrncy counter.

nInstance	index of the device.
nChannel	index of the channel.
nResolution	counter resolution.
nEdge	signal edge.
nEachPeriod	0= after zero crossing, 1 = starts immediately.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_GetAvailableCounterModes( uint
nInstance, uint nChannel, uint* eCOUNTERMODES );
```

gets a counter’s avaliable modes.

nInstance	index of the device.
nChannel	index of the channel.
eCOUNTERMODES	[out] available modes (flags)
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_GetChannelListSize( uint nInstance,
uint* nCLSize );
```

returns the length of the CT measurement channel list.

nInstance	index of the device.
nCLSize	[out] CT measuement channel count.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_GetCounterMode( uint nInstance, uint
nChannelListEntry, uint* eCOUNTERMODE );
```

gets a counter’s avaliable modes.

nInstance	index of the device.
nChannelListEntry	index in channel list.
eCOUNTERMODE	[out] counter mode
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_GetCounterMode_SingleValueMode( uint
nInstance, uint nChannelNumber, uint*
eCOUNTERMODE );
```

nInstance	…
nChannelNumber	…
eCOUNTERMODE	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_GetCounterParameter( uint nInstance,
uint nChannelListEntry, uint eCOUNTERPARAMETER,
uint* nParameterValue );
```

gets a counter’s parameter.

nInstance	index of the device.
nChannelListEntry	index of the channel.
eCOUNTERPARAMETER	parameter index
nParameterValue	[out] parameter value
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_GetCounterParameterBoundaries( uint
nInstance, uint eCOUNTERPARAMETER, uint*
nMinValue, uint* nMaxValue );
```

gets die buondaries of a counter parameter.

nInstance	index of the device.
eCOUNTERPARAMETER	…
nMinValue	[out] minimum value
nMaxValue	[out] maximum value
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_GetCounterParametersForMode( uint
nInstance, uint eCOUNTERMODE, uint*
AvailableCounterParameters );
```

gets the parameters used by a counter mode.

nInstance	index of the device.
eCOUNTERMODE	index of the counter mode.
AvailableCounterParameters	[out] available parameter’s flags
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_GetCounterParameter_SingleValueMode(
uint nInstance, uint nChannelNumber, uint
eCOUNTERPARAMETER, int* nParameterValue );
```

nInstance	…
nChannelNumber	…
eCOUNTERPARAMETER	…
nParameterValue	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_GetFrequency( uint nInstance, double* fFrequency );
```

returns the AD samplerate.

nInstance	index of the device.
fFrequency	[out] AD samplerate in Hz.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_GetIncrementalCounterDirections( uint
nInstance, uint* nDirectionValue );
```

gets the count directions of a IcrceX card.

nInstance	index of the device.
nDirectionValue	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_GetNumberOfValues( uint nInstance, uint*
nNumberOfValues );
```

returns number of measured bursts.

nInstance	index of the device.
nNumberOfValues	[out] number of bursts.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_GetStreamPipeLevel( uint nInstance,
double* fStreamPipeLevel );
```

returns fifo fill level.

nInstance	index of the device.
fStreamPipeLevel	[out] fifo fill level in percent.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_ImpulseCounter( uint nInstance, uint
nChannel, uint nPresetValue, uint nUpDown, uint
nEdge );
```

configures this channel as an impluse counter.

nInstance	index of the device.
nChannel	index of the channel.
nPresetValue	initial counter value.
nUpDown	0= up, 1 = down.
nEdge	signal edge.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_IncrementalCounter( uint nInstance, uint
nChannel, uint nResolution, uint nEdge, uint
nSoftwareReset, uint nHardwareReset, uint
nHWResetEdge, uint nInterpolation );
```

configures this channel as an rotary counter.

nInstance	index of the device.
nChannel	index of the channel.
nResolution	counter resolution.
nEdge	signal edge.
nSoftwareReset	1 = resets counter on measurement start.
nHardwareReset	HW reset mode.
nHWResetEdge	HW reset edge.
nInterpolation	interpolation mode.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_IncrementalCounterExtensionFlow( uint
nInstance, uint nChannel );
```

nInstance	…
nChannel	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_IncrementalCounterExtensionTime( uint
nInstance, uint nChannel, uint nResolution );
```

nInstance	…
nChannel	…
nResolution	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_PeriodCounter( uint nInstance, uint
nChannel, uint nResolution, uint nEdge, uint
nEachPeriod );
```

configures this channel as an period counter.

nInstance	index of the device.
nChannel	index of the channel.
nResolution	counter resolution.
nEdge	signal edge.
nEachPeriod	0= after zero crossing, 1 = starts immediately.
return value	If no error occurs the value "*0*" is returned, otherwise a sub-zero error code.

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```
long CT_PulswidthCounter( uint nInstance, uint
nChannel, uint nResolution, uint nEdge, uint
nEachPeriod );
```

configures this channel as an pulswidth counter.

nInstance	index of the device.
nChannel	index of the channel.
nResolution	counter resolution.
nEdge	signal edge.
nEachPeriod	0= after zero crossing, 1 = starts immediately.
return value	If no error occurs the value "*0*" is returned, otherwise a sub-zero error code.

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```
long CT_ReadData( uint nInstance, uint
nChannelListEntry, uint[] pBuffer, uint*
nNumberOfValues, uint bPartBlock, uint bSkipData
);
```

returns measure data in binary format.

nInstance	index of the device.
nChannelListEntry	index of the channel in the measurement channel list.
pBuffer	[out] counter data.
nNumberOfValues	[in,out] number of values (to) read.
bPartBlock	toggles, if blocks smaller than nNumberOfValues will be read.
bSkipData	toggles, if the data gets transferred or is omitted.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_ReadDataExt( uint nInstance, uint
nChannelListEntry, uint[] pBuffer, uint*
nNumberOfValues, uint bPartBlock, __int64*
AODateTimeT0, double* AODateTimeDeltaT );
```

nInstance	…
nChannelListEntry	…
pBuffer	…
nNumberOfValues	…
bPartBlock	…
AODateTimeT0	…
AODateTimeDeltaT	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_ReadSingleValue( uint nInstance, uint
nChannel, uint* SingleValue );
```

read single raw AD value.

nInstance	index of the device.
nChannel	index of the channel.
SingleValue	[out] counter value.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_Reset( uint nInstance, uint
nHardwareChannelNumber );
```

resets the counter to zero.

nInstance	index of the device.
nHardwareChannelNumber	index of the channel.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_ScaleData( uint nInstance, uint
nChannelListEntry, int[] pInBuffer, double[]
pOutBuffer, uint nNumberOfValues, int&
ARFirstLastValue );
```

nInstance	…
nChannelListEntry	…
pInBuffer	…
pOutBuffer	…
nNumberOfValues	…
ARFirstLastValue	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_SetCounterMode( uint nInstance, uint
nChannelListEntry, uint eCOUNTERMODE );
```

gets a counter’s avaliable modes.

nInstance	index of the device.
nChannelListEntry	index in channel list.
eCOUNTERMODE	counter mode
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_SetCounterMode_SingleValueMode( uint
nInstance, uint nChannelNumber, uint
eCOUNTERMODE );
```

nInstance	…
nChannelNumber	…
eCOUNTERMODE	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_SetCounterParameter( uint nInstance,
uint nChannelListEntry, uint eCOUNTERPARAMETER,
uint nParameterValue );
```

sets counter’s resolution.

nInstance	index of the device.
nChannelListEntry	…
eCOUNTERPARAMETER	…
nParameterValue	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_SetCounterParameter_SingleValueMode(
uint nInstance, uint nChannelNumber, uint
eCOUNTERPARAMETER, int nParameterValue );
```

gets a counter’s avaliable modes.

nInstance	index of the device.
nChannelNumber	…
eCOUNTERPARAMETER	…
nParameterValue	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_SetFrequency( uint nInstance, double
fFreqency );
```

sets the CT samplerate.

nInstance	index of the device.
fFreqency	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_StartMeasure( uint nInstance );
```

starts the CT measurement.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_StopMeasure( uint nInstance );
```

stops the CT measurement.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long CT_UpDownCounter( uint nInstance, uint
nChannel, uint nPresetValue, uint nUpDown, uint
nEdge, uint nSoftwareReset );
```

configures this channel as an up/down counter.

nInstance	index of the device.
nChannel	index of the channel.
nPresetValue	initial counter value.
nUpDown	0= up, 1 = down.
nEdge	signal edge.
nSoftwareReset	1 = resets counter on measurement start.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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Funktionen der Treiber-DLL “MAIv2DLL.dll”



TTL functions.

for controlling and measuring TTL Channels these functions are available:

TTL_ClearChannelList()

TTL_AddChannelToList()

TTL_GetChannelListSize()

TTL_SetFrequency()

TTL_SetChannelDirection()

TTL_GetChannelDirection()

TTL_GetIsChannelDirectionFixed()

TTL_ReadData()

TTL_ReadData8()

TTL_ReadData16()

TTL_ReadData32()

TTL_ReadData64()

TTL_GetBurstSize()

TTL_ReadDataExt()

TTL_ReadPort()

TTL_WritePort()

TTL_ReadBit()

TTL_WriteBit()

TTL_GetStreamPipeLevel()

TTL_GetNumberOfValues()

TTL_InitializeTargetSettings()

TTL_StartMeasure()

TTL_StopMeasure()

TTL_GetPortInfos()

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```
long TTL_AddChannelToList( uint nInstance, uint
nChannel, bool bToADList, uint nDigitalType );
```

adds one channel to the TTL measurement channel list.

nInstance	device Index
nChannel	channel Index
bToADList	if [true], the measurement channel is added to the AD channel list
nDigitalType	width of the channel
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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TTL_ClearChannelList

```
long TTL_ClearChannelList( uint nInstance );
```

clears the TTL measurement channel list.

nInstance	Device Index
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long TTL_GetBurstSize( uint nInstance, uint* nBurstSizeInBytes );
```

gets the burst size of the TTL data in bytes.

nInstance	device Index
nBurstSizeInBytes	burst size of the TTL data
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long TTL_GetChannelDirection( uint nInstance,
uint nChannel, uint* nDirection );
```

returns the diretion (input/output) of a TTL channel.

nInstance	device Index
nChannel	channel Index
nDirection	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long TTL_GetChannelListSize( uint nInstance,
uint* nCLSize );
```

returns the number of TTL measurement channels.

nInstance	Device Index
nCLSize	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long TTL_GetIsChannelDirectionFixed( uint
nInstance, uint nChannel, uint* nDirectionFixed
);
```

indicates, if the diretion (input/output) of a TTL channel is configureable.

nInstance	device Index
nChannel	channel Index
nDirectionFixed	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long TTL_GetNumberOfValues( uint nInstance,
uint* nNumberOfValues );
```

returns number of measured bursts.

nInstance	index of the device.
nNumberOfValues	[out] number of bursts.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long TTL_GetPortInfos( uint nInstance, uint*
NumberOfPorts, uint* PortWidth, uint*
NumberOfDigIn, uint* NumberOfDigOut );
```

returns the properities of a TTL port.

nInstance	device Index.
NumberOfPorts	[out] number of TTL ports.
PortWidth	[out] width (in Bit) of one port.
NumberOfDigIn	[out] number of input channels.
NumberOfDigOut	[out] number of output channels.
return value	If no error occurs the value "*0*" is returned, otherwise a sub-zero error code.

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```
long TTL_GetStreamPipeLevel( uint nInstance,
double* fStreamPipeLevel );
```

returns fifo fill level.

nInstance	index of the device.
fStreamPipeLevel	[out] fifo fill level in percent.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long TTL_InitializeTargetSettings( uint
nInstance );
```

nInstance	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long TTL_ReadBit( uint nInstance, uint nChannel,
uint* nBit );
```

returns the state of a TTL channel.

nInstance	device Index
nChannel	…
nBit	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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TTL_ReadData

```
long TTL_ReadData( uint nInstance, uint[]
pBuffer, uint* nNumberOfBursts, uint bPartBlock,
uint bSkipData );
```

reads the TTL measurement values.

nInstance	device Index
pBuffer	measurement value buffer
nNumberOfBursts	…
bPartBlock	unused.
bSkipData	unused.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long TTL_ReadData16( uint nInstance, ushort*
pBuffer, uint* nNUmberOfBursts );
```

reads the TTL measurement values as a 16 Bit array.

nInstance	device Index
pBuffer	measurement value buffer
nNUmberOfBursts	[in,out] in = number of values (to) read
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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TTL_ReadData32

```
long TTL_ReadData32( uint nInstance, uint*
pBuffer, uint* nNUmberOfBursts );
```

reads the TTL measurement values as a 32 Bit Array.

nInstance	device Index
pBuffer	measurement value buffer
nNUmberOfBursts	[in,out] in = number of values (to) read
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long TTL_ReadData64( uint nInstance, u64*
pBuffer, uint* nNUmberOfBursts );
```

reads the TTL measurement values as a 64 bit array.

nInstance	device Index
pBuffer	measurement value buffer
nNUmberOfBursts	[in,out] in = number of values (to) read
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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TTL_ReadData8

```
long TTL_ReadData8( uint nInstance, uchar*
pBuffer, uint* nNUmberOfBursts );
```

reads the TTL measurement values as an 8 bit array.

nInstance	device Index
pBuffer	measurement value buffer
nNUmberOfBursts	[in,out] in = number of values (to) read
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long TTL_ReadDataExt( uint nInstance, uchar*
pBuffer, uint* nNUmberOfBursts, uint
nBurstSizeInBytes, __int64* AODateTimeT0,
double* AODateTimeDeltaT );
```

reads the TTL measurement values.

nInstance	device Index
pBuffer	measurement value buffer
nNUmberOfBursts	[in,out] in = number of values (to) read
nBurstSizeInBytes	[in] burst size of the TTL data in bytes
AODateTimeT0	[out] T0 of the first burst
AODateTimeDeltaT	[out] delta T between the bursts
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long TTL_ReadPort( uint nInstance, uint nPort,
uint* nPortValue );
```

returns the state of a TTL port.

nInstance	device Index
nPort	index of the port to read
nPortValue	[out] the port’s state
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long TTL_SetChannelDirection( uint nInstance,
uint nChannel, uint nDirectionValue );
```

sets the diretion (input/output) for a TTL channel.

nInstance	device Index
nChannel	channel Index
nDirectionValue	direction: Input = 0, output = 1
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long TTL_SetFrequency( uint nInstance, double
fFrequency );
```

sets a new TTL sample rate.

nInstance	device Index
fFrequency	samplerate
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long TTL_StartMeasure( uint nInstance );
```

starts the TTL measurement.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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TTL_StopMeasure

```
long TTL_StopMeasure( uint nInstance );
```

stops the TTL measurement.

nInstance	index of the device.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long TTL_WriteBit( uint nInstance, uint
nChannel, uint nBit );
```

sets a new state to a TTL channel.

nInstance	device Index
nChannel	…
nBit	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long TTL_WritePort( uint nInstance, uint nPort,
uint nPortValue );
```

writes a new state to a TTL port.

nInstance	device Index
nPort	index of the port to read
nPortValue	the port’s state
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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Funktionen der Treiber-DLL “MAIv2DLL.dll”



PWM functions.

for controlling and measuring PWM Channels these functions are available:

PWM_StartOutput()

PWM_StopOutput()

PWM_PreparePWMOutput()

PWM_PrepareFMOutput()

PWM_WriteNewPWMRatio()

PWM_WriteNewFMFrequency()

PWM_GetPWMRatio()

PWM_GetFMFrequency()

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```
long PWM_GetFMFrequency( uint nInstance, uint
nChannel, uint* Frequency );
```

returns the frequency of the FM output.

nInstance	device index.
nChannel	channel index.
Frequency	[out] frequency.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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PWM_GetPWMRatio

```
long PWM_GetPWMRatio( uint nInstance, uint
nChannel, uint* nRatio );
```

returns the pulse/pause ratio of the PWM output.

nInstance	device index.
nChannel	channel index.
nRatio	[out] pulse/pause ratio.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long PWM_PrepareFMOutput( uint nInstance, uint
nChannel, uint StartFrequency, uint
nPulsWidthRatio );
```

configures the FM output.

nInstance	device index.
nChannel	channel index.
StartFrequency	initial frequency.
nPulsWidthRatio	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long PWM_PreparePWMOutput( uint nInstance, uint
nChannel, uint StartFrequency, uint nStartRatio
);
```

configures the PWM output.

nInstance	device index.
nChannel	channel index.
StartFrequency	frequency.
nStartRatio	initial pulse/pause ratio.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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PWM_StartOutput

```
long PWM_StartOutput( uint nInstance );
```

starts the PWM output.

nInstance	device index.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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PWM_StopOutput

```
long PWM_StopOutput( uint nInstance );
```

stops the PWM output.

nInstance	device index.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long PWM_WriteNewFMFrequency( uint nInstance,
uint nChannel, uint Frequency );
```

changes the frequency of the FM output.

nInstance	device index.
nChannel	channel index.
Frequency	new frequency.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long PWM_WriteNewPWMRatio( uint nInstance, uint
nChannel, uint nRatio );
```

changes the pulse/pause ratio of the PWM output.

nInstance	device index.
nChannel	channel index.
nRatio	new pulse/pause ratio.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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Funktionen der Treiber-DLL “MAIv2DLL.dll”



SSI functions.

for controlling and measuring SSI Channels these functions are available:

SSI_SetBitWidth()

SSI_ConfigSSI()

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```
long SSI_ConfigSSI( uint nInstance, uint nReset,
uint nResetEdge, uint nActivePassiveMode, uint
nClockFrequency, uint nEdge );
```

configures the SSI port.

nInstance	device index.
nReset	if [1] = reset signal is enabled.
nResetEdge	reset signal edge.
nActivePassiveMode	active/passive mode.
nClockFrequency	clock frequency.
nEdge	signal edge.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long SSI_SetBitWidth( uint nInstance, uint
nBitWidth );
```

sets the width (in bits) of the SSI Port.

nInstance	index of the device.
nBitWidth	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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Funktionen der Treiber-DLL “MAIv2DLL.dll”



online functions.

for DSP cards, which have installed the corresponding online functions, these functions are available:

Realtime_AddTriggerEntry()

Realtime_AddFilterEntry()

Realtime_DesignIIRFilter()

Realtime_DesignFIRFilter()

Realtime_AddPIDEntry()

Realtime_ChangePID_Params()

Realtime_ChangePID_NominalValue()

Realtime_AddFFTEntry()

Realtime_FFT_ReadData()

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```
long Realtime_AddFFTEntry( uint nInstance, uint
nChannelListEntry, uint nAnzahlPunkte, uint
nSpektrum, double dGain, uint nFensterFunktion,
double dParameter, uint* AOIndexInChannelsRTList
);
```

adds an online FFT analyzer.

nInstance	device index.
nChannelListEntry	index of the AD channel in the measurement channel list.
nAnzahlPunkte	number of points in the spectrum.
nSpektrum	type of spectrum.
dGain	Gain factor
nFensterFunktion	FIR window Function.
dParameter	optional parameter for some window functions.
AOIndexInChannelsRTList	[out] index of this functions’s entry inside the online function list.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long Realtime_AddFilterEntry( uint nInstance,
uint AIChannelListEntry, uint AIFilterMode, uint
AIFilterOrder, double[] AICoefficients, double
AIOutputGainFactor, uint*
AOIndexIndChannelsRTList );
```

adds an online filter.

nInstance	device index.
AIChannelListEntry	index of the AD channel in the measurement channel list.
AIFilterMode	internal filter type index.
AIFilterOrder	filter order.
AICoefficients	[in] filter coefficients.
AIOutputGainFactor	Gain of the Filters.
AOIndexIndChannelsRTList	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long Realtime_AddPIDEntry( uint nInstance, uint
nChannelListEntry, double dPAnteil, double
dIAnteil, double dDAnteil, uint nReglerArt,
double Gain, double dSollwert, double dMinValue,
double dMaxValue, uint nDAKanal, double
nSampleFreq, uint* AOIndexInChannelsRTList );
```

adds an online PID controller.

nInstance	device index.
nChannelListEntry	index of the AD channel in the measurement channel list.
dPAnteil	P portion.
dIAnteil	I portion.
dDAnteil	D portion.
nReglerArt	type of PID controller to create.
Gain	Gain factor
dSollwert	desired value.
dMinValue	minimum value.
dMaxValue	maximum value.
nDAKanal	index des DA channel to be used for output.
nSampleFreq	sample rate (use the same as given to AD_SetSampleRate).
AOIndexInChannelsRTList	[out] index of this functions's entry inside the online function list.
return value	If no error occurs the value "*0*" is returned, otherwise a sub-zero error code.

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Funktionen der Treiber-DLL “MAIv2DLL.dll”

Realtime_AddTriggerEntry



```
long Realtime_AddTriggerEntry( uint nInstance,
uint nChannelListEntry, uint nStartTriggerAktiv,
double dStartSchwellwert1, double
dStartSchwellwert2, uint nStartTriggerTyp, uint
nStartEnableAtStart, uint nStartTriggerMessung,
uint nStartTriggerSetTTL, uint
nStartTriggerTTLKlemme, uint
nStartTriggerTTLZustand, uint
nStartTriggerTTLFlanke, uint nStopTriggerAktiv,
double dStopSchwellwert1, double
dStopSchwellwert2, uint nStopTriggerTyp, uint
nStopEnableAtStart, uint nStopTriggerMessung,
uint nStopTriggerSetTTL, uint
nStopTriggerTTLKlemme, uint
nStopTriggerTTLZustand, uint
nStopTriggerTTLFlanke, uint nRetriggerbar, uint*
AOIndexInChannelsRTList );
```

adds an online trigger.

nInstance	device index.
nChannelListEntry	index of the AD channel in the measurement channel list.
nStartTriggerAktiv	if [1]= start trigger is enabled.
dStartSchwellwert1	threshold 1 for the start trigger.
dStartSchwellwert2	threshold 2 for the start trigger.
nStartTriggerTyp	type of the start trigger.
nStartEnableAtStart	if [1] = start trigger is armed at measurement start.
nStartTriggerMessung	if [1] = start trigger starts AD measurement.
nStartTriggerSetTTL	if [1] = start trigger sets a TTL channel’s state.
nStartTriggerTTLKlemme	index of the start trigger’s TTL channel.

nStartTriggerTTLZustand	initial state of the start trigger's TTL channel.
nStartTriggerTTLFlanke	triggered state of the start trigger's TTL channel.
nStopTriggerAktiv	if [1]= stop trigger is enabled.
dStopSchwellwert1	threshold 1 for the stop trigger.
dStopSchwellwert2	threshold 2 for the stop trigger.
nStopTriggerTyp	type of the start trigger.
nStopEnableAtStart	if [1] = stop trigger is armed at measurement start.
nStopTriggerMessung	if [1] = stop trigger starts AD measurement.
nStopTriggerSetTTL	if [1] = stop trigger sets a TTL channel's state.
nStopTriggerTTLKlemme	index of the stop trigger's TTL channel.
nStopTriggerTTLZustand	initial state of the stop trigger's TTL channel.
nStopTriggerTTLFlanke	triggered state of the stop trigger's TTL channel.
nRetriggerbar	if [1] = triggers can arm each other.
AOIndexInChannelsRTList	[out] index of this functions's entry inside the online function list.
return value	If no error occurs the value "*0*" is returned, otherwise a sub-zero error code.

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```
long Realtime_ChangePID_NominalValue( uint
nInstance, uint nChannelListEntry, uint
nIndexInChannelsRTList, double dSollwert );
```

changes an online PID controller’s nominal value.

nInstance	device index.
nChannelListEntry	index of the AD channel in the measurement channel list.
nIndexInChannelsRTList	…
dSollwert	nominal value.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long Realtime_ChangePID_Params( uint nInstance,
uint nChannelListEntry, uint
nIndexInChannelsRTList, double dPAnteil, double
dIAnteil, double dDAnteil );
```

Changes an online PID controller’s parameters.

nInstance	device index.
nChannelListEntry	index of the AD channel in the measurement channel list.
nIndexInChannelsRTList	…
dPAnteil	P portion.
dIAnteil	I portion.
dDAnteil	D portion.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long Realtime_DesignFIRFilter( double AISampleRate,
uint AIFilterType, double AICutoffFreqLow, double
AICutoffFreqHigh, bool AIFrequenciesAreNormalized,
uint AIOrder, uint AIFilterWindow, double
AIWindowParameter, bool AIScalePassband, uint*
AOFilterMode, double[] AOCoefficientsList );
```

designs an FIR filter.

AISampleRate	Sample Rate (use the same as given to AD_SetSampleRate)..
AIFilterType	type of Filter to design (Lowpass/Highpass/Bandstop/BandPass).
AICutoffFreqLow	lower cutoff freqency.
AICutoffFreqHigh	upper cutoff freqency.
AIFrequenciesAreNormalized	if [true] the given values for frequencies will be treated as already normalized values.
AIOrder	filter order.
AIFilterWindow	window function to use.
AIWindowParameter	optional parameter for some window functions.
AIScalePassband	if [true] = the algorithm tries to scale the passband to 1.
AOFilterMode	[out] internal filter type index.
AOCoefficientsList	[out] list of filter coefficients.
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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Realtime_DesignIIRFilter

```
long Realtime_DesignIIRFilter( double AISampleRate,
uint AIFilterType, double AICutoffFreqency, double
AICutoffFreqency2, bool AIFrequenciesAreNormalized,
double AIPassabsorbtion, double AIStopbandattenuation,
uint AIDesignMethod, uint AIOrder, uint* AOFilterMode,
double[] AOCoefficientsList, double* AOGain );
```

designs an IIR Filter.

AISampleRate	Sample Rate (use the same as given to AD_SetSampleRate).
AIFilterType	Type of Filter to design (Lowpass/Highpass/Bandstop/BandPass).
AICutoffFreqency	lower cutoff freqency.
AICutoffFreqency2	upper cutoff freqency.
AIFrequenciesAreNormalized	if [true] the given values for frequencies will be treated as already normalized values.
AIPassabsorbtion	passband absorbtion.
AIStopbandattenuation	stopband attenuation.
AIDesignMethod	design method to use.
AIOrder	filter order.
AOFilterMode	[out] internal filter type index.
AOCoefficientsList	[out] list of filter coefficients.
AOGain	…
return value	If no error occurs the value “*0*” is returned, otherwise a sub-zero error code.

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```
long Realtime_FFT_ReadData( uint nInstance, uint
nChannelListEntry, double[] FFTData, uint*
nNumberOfValues );
```

returns measure data in binary format.

nInstance	index of the device.
nChannelListEntry	index of the AD channel in the measurement channel list.
FFTData	[out] FFT spectrum.
nNumberOfValues	[in,out] number of values (to) read.
return value	If no error occurs the value "*0*" is returned, otherwise a sub-zero error code.

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Fehlercodes, welche während einer Messung auftreten können, sowie ihre Bedeutung.

-1000	Die Messkarte ist nicht vorhanden oder die Nummer der Karte ist ungültig.
-1010	Es konnte nicht von der Messkarte gelesen werden.
-1020	Es konnte nicht auf die Messkarte geschrieben werden.
-1030	Die Klemmstelle ist ungültig oder der Kanal nicht vorhanden.
-1040	Der eingegebene Parameter ist falsch.
-1050	Das Löschen aller Kanallisten ist fehlgeschlagen.
-1060	Das Löschen der AD-Kanalliste ist fehlgeschlagen.
-1070	Das Löschen der TTL-Kanalliste ist fehlgeschlagen.
-1080	Das Löschen der CT-Kanalliste ist fehlgeschlagen.
-1090	Das Löschen der DA-Kanalliste ist fehlgeschlagen.
-1100	Fehler beim hinzufügen eines AD-Kanals in die Kanalliste.
-1110	Fehler beim hinzufügen eines TTL-Kanals in die Kanalliste.
-1120	Fehler beim hinzufügen eines CT-Kanals in die Kanalliste.
-1130	Fehler beim hinzufügen eines DA-Kanals in die Kanalliste.
-1140	Fehler bei der manuellen Analogausgabe.
-1150	Die eingestellte Frequenz ist zu niedrig.
-1160	Die eingestellte Frequenz ist zu hoch.
-1170	Die Direction kann nicht verändert werden.
-1180	Die Konfiguration vor dem Begin der Messung war nicht erfolgreich.
-1190	Die Konfiguration vor dem Begin der Messung war nicht erfolgreich.
-1200	Die Konfiguration vor dem Begin der Messung war nicht erfolgreich.

-1210 Die Konfiguration vor dem Begin der Messung war nicht erfolgreich.
-1220 Die Messung wurde nicht gestartet.
-1230 Die Messung wurde nicht gestoppt.
-1240 Die AD-Datenausgabe ist fehlgeschlagen.
-1250 Die TTL-Datenausgabe ist fehlgeschlagen.
-1260 Die CT-Datenausgabe ist fehlgeschlagen.
-1270 Die DA-Datenausgabe ist fehlgeschlagen.
-1280 Während des Setzens der DA-Daten trat ein Fehler auf.
-1290 Das SoftwareFIFO ist übergelaufen. Bitte Paketgröße überprüfen.
-1300 Fehler bei der Parametrierung des Impulsezählers.
-1310 Fehler bei der Parametrierung des Frequenzzählers.
-1320 Fehler bei der Parametrierung des Periodendauerzählers.
-1330 Fehler bei der Parametrierung des Pulsweitenzählers.
-1340 Fehler bei der Parametrierung des Inkrementalzählers.
-1350 Fehler beim Setzen der Ausgabefrequenz.
-1360 Unerlaubter Zugriff.
-1370 Ungültiger Zeiger (Pointer).
-1380 Fehler beim Starten der analogen Ausgabe.
-1390 Fehler beim Stoppen der analogen Ausgabe.

-1400 Es wurde keine automatische Messung / Ausgabe konfiguriert. Der Start wurde abgebrochen.
-1410 Fehler beim Setzen der Frequenz für eine Zaehlermessung.
-1420 Fehler beim Setzen der Frequenz für eine analoge Messung.

-1500 Mehr als eine Karte wurde auf Master gestellt.
-1510 Keine Karte wurde auf Master gestellt.
-1520 Eine Karte, welche auf Master eingestellt ist, kann auf kein externes Ereignis triggern.
-1530 Im angewaehltem Differenzmessmodus steht dieser AD Kanal nicht zur Verfuegung.

-1600 Eine AD-Messung ist mit dieser Karte nicht möglich.
-1610 Eine TTL-Messung ist mit dieser Karte nicht möglich.
-1620 Eine CT-Messung ist mit dieser Karte nicht möglich.
-1630 Eine DA-Ausgabe ist mit dieser Karte nicht möglich.
-1640 Diese Karte hat keine digitalen Kanäle.
-1650 Diese Funktion wird von der Karte aktuell nicht, bzw. noch nicht unterstuetzt.

-xxxx Unbekannter Fehler.